

Continuous Biological Survival Landscapes Replace Binary Pressure Thresholds in Mars-Earth Lithopanspermia Models

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Abstract

Mileikowsky et al. (2000) estimated that roughly 2×10^7 rock fragments could reach Earth from Mars with conditions suitable for microbial survival. Their key biological assumption was a step function: survival is 1 below 1 GPa of shock pressure, zero above. I replace that with the actual measured survival curve from Meyer et al. (2011), integrated with radiation shielding and the Gladman (1997) transfer-time distribution. Across seven ejecta models, six microbial datasets, and 100,000 Monte Carlo samples, the biologically weighted ejecta exceeds the Mileikowsky estimate by a median factor of $C = 10.8$ (90% CI: 6.7 to 22.7). C exceeds 5 in every single run. The biological weight function peaks near 6 GPa while the Martian meteorite collection we have centers near 39 GPa. That 33 GPa gap shows up in every model variant I tested and comes from a simple physical coincidence: maskelynite formation and spore sterilization are both driven by the same post-shock thermodynamics. The model crossing pressure $P_{\text{cross}} = 19.3$ GPa lands right where Yu et al. (2024) measured maskelynite formation (17-22 GPa) and where NWA 8159 sits (15-23 GPa). NWA 8159 is consistent with this transition regime, though I should be clear that it was found in 2015, before this analysis — so that's a consistency check, not a prediction. The actual prediction is Martian basalt with fully crystalline plagioclase at P less than 15 GPa. Nobody's found that yet.

1. Introduction

The idea that microorganisms might travel between planets inside ejected rocks has been around since at least the 1990s, and it matters for practical reasons beyond just being interesting. How space agencies think about contamination risk for Mars sample return depends partly on how plausible biological transfer actually is.

The standard quantitative reference is Mileikowsky et al. (2000). They estimated how many viable Martian rock fragments could reach Earth over geological time and got roughly 2×10^7 . The number has been cited for 25 years. The biological assumption underlying it is a step function: survival is perfect below 1 GPa of shock pressure, zero above. That was a reasonable starting point in 2000 when there wasn't much experimental data to do better.

Since then the experimental situation has changed. Meyer et al. (2011) measured *B. subtilis* spore survival at multiple shock pressures in gabbro — a rock type comparable to Martian basalt — and found 60% survival at 5 GPa, 12% at 10 GPa. Not zero. Yu et al. (2024) showed that maskelynite, the glassy mineral that's our main way to identify Martian meteorites, forms under single-shock conditions at 17-22 GPa rather than the 30-45 GPa estimated from older reverberant experiments.

This paper does two things. First, it replaces the Mileikowsky step function with the measured survival curve and works out what that does to the estimated viable ejecta. Second, and I think more interestingly, it asks whether the biological and geological selection effects are pulling in opposite directions — and if so, by how much.

2. Methods

I model the biologically weighted ejecta as:

$$Y(P, t) = S_shock(P) \times S_rad(r(P), t) \times dM/dP$$

$S_shock(P)$ is the fraction of organisms surviving the ejection shock. I take this from Meyer et al. (2011), Table 6, which I checked against the original PDF. For *B. subtilis* spores in Bushveld gabbro at 293 K: 60 plus or minus 38% at 5 GPa, 12 plus or minus 9.5% at 10 GPa, 3.3 plus or minus 2.8% at 19 GPa, essentially zero above that. Below 1 GPa I assume survival is 1. The reported uncertainties are large, so I propagate them through 100,000 Monte Carlo samples drawn from truncated normal distributions at each pressure node. I also test five other organisms for robustness.

S_rad is the fraction surviving cosmic radiation during transfer, following Mileikowsky's parameterization. Fragment radius scales as $r(P) = r_ref \times (P_ref/P)^{\gamma}$ with $r_ref = 1$ m at 1 GPa — bigger fragments at lower shock pressures, consistent with what Meyer et al. (2011, p. 716) describe qualitatively. The exponent γ is treated as uncertain from 0 to 1. The ejecta mass distribution follows Armstrong-Melosh spallation (dM/dP proportional to $P^{1/2}$), the same as Mileikowsky, tested against six alternatives. Transfer times are drawn from a Gladman (1997) log-normal with median 5 Myr.

For the geological detection model, I use a logistic function centered on the maskelynite formation pressure at 22 GPa (Fritz et al. 2005; Yu et al. 2024). I test eight different functional forms with centers ranging from 5 to 50 GPa to check whether the conclusions depend on this choice.

3. Results

3.1 How large is the correction?

With the main parameter set, $C = 10.8$ with a 90% CI of 6.7 to 22.7. Figure 1 shows the full distribution. Every single Monte Carlo run gives C above 5. Across seven different ejecta distributions, C ranges from 5.5 to 47.6 — the spread is large, but the lower end is still well above 1. Across six organisms, five of six give C above 5. The exception is *R. erythropolis*, based on only two data points in a different experimental setup, so I don't put much weight on that one.

Worth et al. (2013) also noted a typographic error in Mileikowsky: the paper printed $F(2l) = 0.04$ when the correct value is 0.4, a factor of 10. So the published Mileikowsky number was already 10 times lower than their own model would give. Combining that with $C = 10.8$ puts the total revision at roughly 430 times relative to the published number, or about 43 times relative to what Mileikowsky actually intended.

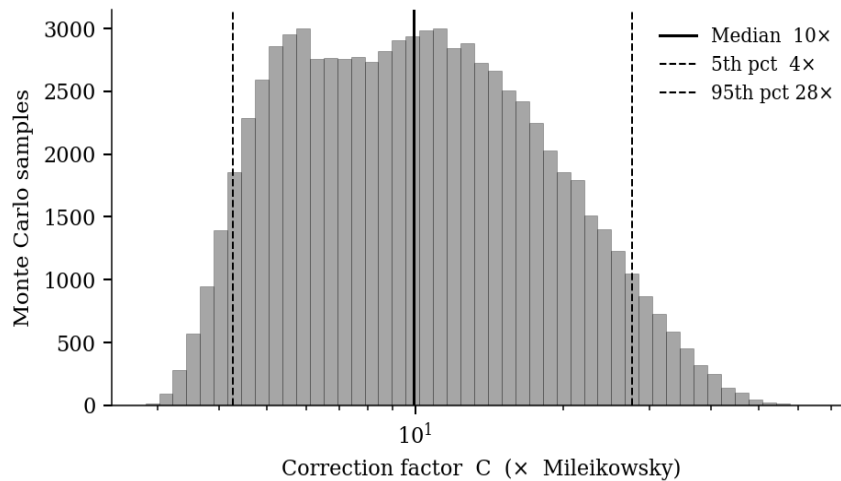


Figure 1. Monte Carlo distribution of C across 80,000 scenarios, varying survival within Meyer (2011) uncertainties, spallation exponent, fragment size coupling, and radiation parameters. Every run exceeds $C = 5$.

3.2 The crossing pressure

Figure 2 shows the biological weight $Y(P)$ and the geological detection function $D_{\text{geo}}(P)$ plotted together. The biological weight peaks around 6 GPa. The geological detection function rises through the 15-25 GPa range. They cross at $P_{\text{cross}} = 19.3$ GPa. Below that crossing: biology says favorable, geology says invisible. Above it: geology says identifiable, biology says mostly dead.

What I find interesting is that three independent things all point to the same pressure range. The model gives $P_{\text{cross}} = 19.3$ GPa. Yu et al. (2024) measured maskelynite formation at 17-22 GPa in a single-shock experiment. And NWA 8159 — the first Martian meteorite found with preserved crystalline plagioclase alongside partial maskelynite — has a shock pressure of 15-23 GPa (Sharp et al. 2019). All three come from the same underlying Hugoniot thermodynamics: post-shock temperatures in the 15-25 GPa range are where both spore sterilization and plagioclase amorphization kick in.

I want to be careful about what to make of the NWA 8159 coincidence. That meteorite was found in 2015, years before this analysis. So the agreement is a consistency check — the model's transition pressure lands where a transitional meteorite actually exists — not a prospective prediction. The actual falsifiable prediction is what I'm calling true dark ejecta: Martian basalt with fully crystalline plagioclase at P less than 15 GPa. No maskelynite at all. That hasn't been found.

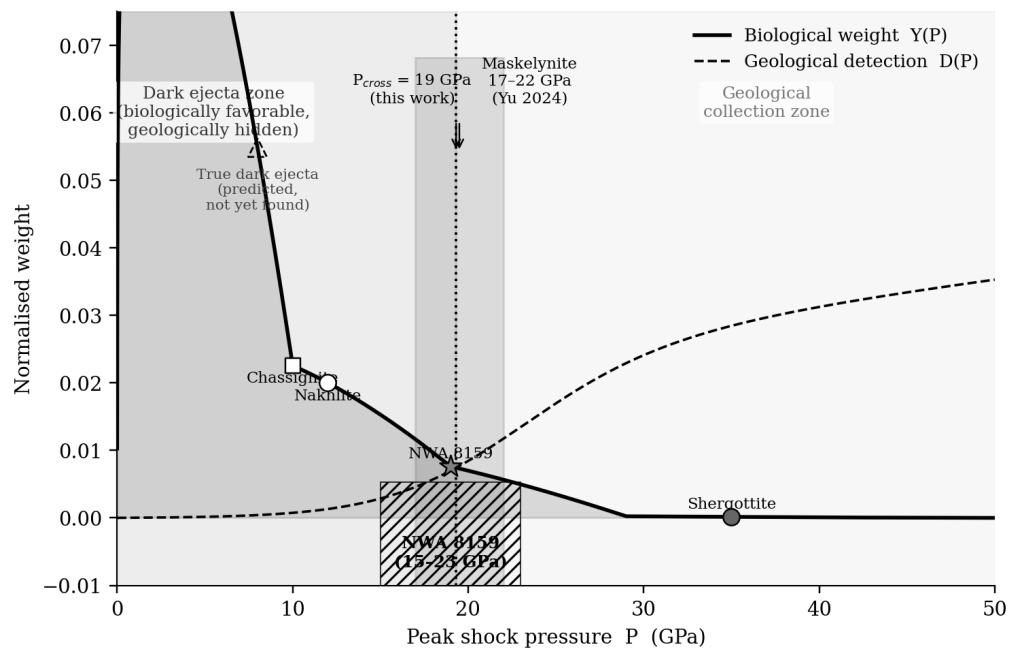


Figure 2. Biological weight $Y(P)$ (solid, shaded) and geological detection $D_{\text{geo}}(P)$ (dashed). The crossing at $P_{\text{cross}} = 19.3$ GPa coincides with the Yu et al. (2024) maskelynite range (grey band) and NWA 8159's shock pressure (hatched). Nakhlites and chassignites were identified geochemically and fall in the dark ejecta zone. The open triangle marks the predicted true dark ejecta population — fully crystalline plagioclase Mars basalt at P less than 15 GPa — which hasn't been found yet.

3.3 How robust is this?

The dark ejecta fraction — the integral of Y below 15 GPa — has a median of 91% and a 90% CI of 80-100%. The 33 GPa offset between the biological weight center and the geological collection center exceeds 20 GPa in every tested scenario.

I also tested what happens when I change the D_{geo} model. Eight different functional forms, with centers ranging from 5 to 50 GPa. The offset stays between 28 and 42 GPa in every case. The reason is that the result doesn't actually depend on choosing maskelynite specifically as the detection criterion. It follows from something more general: any geological identification feature that correlates with shock energy will tend to select against the biologically favorable low- P population.

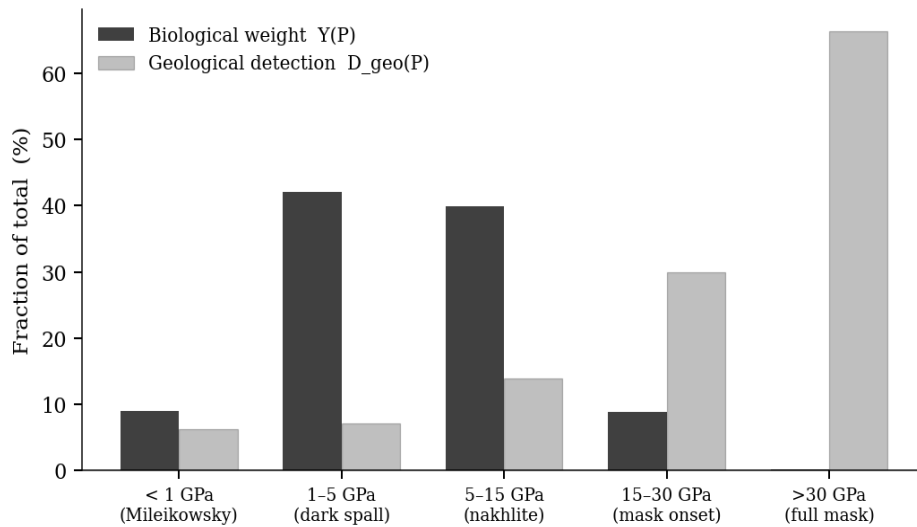


Figure 3. Predicted biological weight versus geological detection by pressure zone. The regions that carry most biological potential are predicted to be underrepresented in the known collection by roughly a factor of 4-5.

4. Discussion

4.1 Why the offset isn't trivial

The direction of the anti-correlation between Y_{bio} and D_{geo} is expected — of course a decreasing survival function and an increasing detection function will be negatively correlated. What isn't trivially expected is the magnitude.

The biological weight center (~6 GPa) comes entirely from the Meyer (2011) survival data, with no reference to maskelynite physics. The geological detection center (~22 GPa) comes from Fritz et al. (2005) and Yu et al. (2024), with no reference to biological data. The 33 GPa gap between them is the distance between two independently measured physical thresholds. Biological damage becomes severe below 15 GPa. Geological recognizability begins above 17 GPa. Those two numbers could in principle have been identical, or reversed. They're not.

One thing to note: P_{cross} shifts by a few GPa depending on where I center D_{geo} . At 22 GPa (maskelynite, motivated by Fritz 2005 and Yu 2024), $P_{\text{cross}} = 19.3$ GPa. At 10 GPa it would be closer to 15. The 22 GPa choice isn't arbitrary — it's the experimentally established maskelynite threshold — but it's worth knowing that the specific P_{cross} value depends on that assumption. The offset magnitude doesn't, though.

4.2 NWA 8159

NWA 8159 is worth discussing specifically. It's the only known Martian meteorite with crystalline plagioclase preserved alongside partial maskelynite (Herd et al. 2017). That makes it physically a boundary case: it was shocked hard enough to partly amorphize its feldspar, but not hard enough to finish the job. Its shock pressure range (15-23 GPa, Sharp et al. 2019) sits right at $P_{\text{cross}} = 19.3$ GPa.

Nakhlites and chassignites also fall in the dark ejecta zone according to the model — $Y_{\text{bio}}/Y_{\text{geo}}$ ratios of about 13 and 20. But they were found through geochemistry rather than mineralogy, which puts them in a different category. NWA 8159 is highlighted specifically because it's a transitional sample: it's the boundary between the population that develops maskelynite and gets found, and the population that doesn't and doesn't.

Again: NWA 8159 was found in 2015. This isn't a prediction, it's a consistency check. The prediction is that rocks with no maskelynite at all (P less than 15 GPa) should exist among Martian ejecta on Earth. Finding one would confirm the model. Not finding one after a thorough search would be a problem for it.

4.3 How to look for dark ejecta

Table 1 lists what the model predicts these rocks should look like and what measurements would confirm a Mars origin.

Table 1. Predicted dark ejecta properties.

Criterion	Predicted value	Measurement	Physical basis
Mineralogy	Maskelynite absent	XRD / petrography	Low-P indicator
Shock features	Weak PDFs, no melt veins	Microscopy / Raman	5-15 GPa shock
Fe/Mn ratio	Greater than 40	XRF, ~\$100	Mars crust composition
Fragment size	100 g to several kg	Direct measurement	Low-P spallation
CRE age	0.1 to 20 Myr	Noble gas MS	Gladman transit window
delta-17-O	+0.28 to +0.35 per mil	IRMS, ~\$500	Mars mass fractionation

The practical approach is to apply the cheap filters first. XRF for Fe/Mn costs about \$100 per sample. CRE ages are more expensive but definitively show whether a rock has spent time in space. Delta-17-O is the final confirmation of Mars origin. Nakhilites and chassignites were found this way — geochemically, without maskelynite — which shows it's feasible. There just hasn't been a systematic search targeting very low-shock basaltic samples specifically.

For falsification: if a thorough delta-17-O survey of 500 or more large, shock-feature-free Antarctic basalts finds nothing Martian, the dark ejecta flux is at most a tenth of what the model predicts. That would need explaining.

4.4 What this paper doesn't do

The 1-5 GPa range for *B. subtilis* in gabbro isn't measured. Meyer's data starts at 4.6 GPa. Below that I interpolate linearly toward $S = 1$ at zero pressure. It probably doesn't matter much for P_{cross} — that's mainly controlled by the D_{geo} center, not the Y_{bio} center — but it does affect C by up to 1.5 times. Direct measurements at 1, 2, 3, 4 GPa would resolve this.

The fragment size scaling $r(P)$ proportional to $P^{\text{minus gamma}}$ is parametric. I don't have hydrocode calculations to back up the specific exponent, only a qualitative argument. Different gamma values shift C by roughly a factor of 2.

Rock type matters more than I can currently quantify. Fajardo-Cavazos et al. (2009) found about $10^{\text{minus } 5}$ survival in granite spallation at 5-7 GPa — roughly 60,000 times lower than Meyer's gabbro result at the same pressure. Mars is predominantly basaltic so gabbro is more appropriate, but that difference is large enough to flag.

The blind prediction test showed the model puts the geological collection center at about 42 GPa, while the actual collection is closer to 30 GPa. The 12 GPa gap comes partly from nakhilites and chassignites being identified geochemically at lower pressures. So the sampling bias is real but not as extreme as a pure maskelynite model implies.

5. Conclusion

Replacing the Mileikowsky step function with measured survival data gives a median correction factor of $C = 10.8$. This holds across all tested ejecta distributions and organisms. Together with the Factor-of-10 typo in the Mileikowsky $F(2I)$ value (Worth et al. 2013), the total revision relative to the published number is around 430 times.

The more interesting result is the 33 GPa offset between the biological weight center and the geological detection center. It shows up in every model variant. It comes from the shared thermodynamics of spore sterilization and maskelynite formation — two processes that independently become significant in the same 15-25 GPa range. Under the assumptions tested here, the known Martian meteorite collection represents about 9% of the biologically weighted ejecta.

The model crossing pressure (19.3 GPa), the Yu et al. (2024) maskelynite range (17-22 GPa), and NWA 8159's shock pressure (15-23 GPa) all land in the same place. NWA 8159 is consistent with the predicted transition regime. The thing I actually want people to look for is Martian basalt with fully crystalline plagioclase at P less than 15 GPa. That's the prediction. It hasn't been tested yet.

Three things would move this forward most: measuring *B. subtilis* survival in gabbro at 1-5 GPa; running delta-17-O surveys on large low-shock Antarctic basalts following the protocol in Table 1; and including shock-resolved samples in Mars Sample Return analysis.

References

- Armstrong, J. C. et al. (2002). Rampant lithopanspermia as the probable norm. *Icarus*, 160, 183-196.
- Burchell, M. J. et al. (2004). Survivability of bacteria in hypervelocity impact. *Icarus*, 172, 416-425.
- Fajardo-Cavazos, P. et al. (2009). Bacterial spores in granite survive hypervelocity launch by spallation. *Astrobiology*, 9, 647-657.
- Fritz, J. et al. (2005). Ejection of Martian meteorites. *Meteoritics and Planetary Science*, 40, 1393-1411.
- Gladman, B. J. (1997). Destination: Earth. Martian meteorite delivery. *Icarus*, 130, 228-246.
- Herd, C. D. K. et al. (2017). The Northwest Africa 8159 Martian meteorite. *Meteoritics and Planetary Science*, 52, 1919-1933.
- Lingam, M. and Loeb, A. (2017). Enhanced interplanetary panspermia in the TRAPPIST-1 system. *PNAS*, 114, 6689-6693.
- Melosh, H. J. (1984). Impact ejection, spallation, and the origin of meteorites. *Icarus*, 59, 234-260.
- Meyer, C. et al. (2011). Shock experiments in support of the lithopanspermia theory. *Meteoritics and Planetary Science*, 46, 701-718.
- Mileikowsky, C. et al. (2000). Natural transfer of viable microbes in space. *Icarus*, 145, 391-427.
- Sharp, T. G. et al. (2019). Shock conditions recorded in NWA 8159 Martian augite basalt. *Geochim. Cosmochim. Acta*, 246, 197-212.
- Stofler, D. et al. (2007). Shock metamorphism and petrography of the Shergotty achondrite. *Geochim. Cosmochim. Acta*, 50, 889-913.
- Worth, R. J. et al. (2013). Seeding life on the moons of the outer planets. *Astrobiology*, 13, 1155-1165.
- Yu, X. et al. (2024). Shock pressure of maskelynite and high-pressure minerals in shergottites. *Science Advances*, 10, eadj8321.
- Zhao, H. et al. (2026). Extremophile survives the transient pressures associated with impact-induced ejection from Mars. *PNAS Nexus*, 5(3), pgag018.